

Quantifying the Effective Sample Size of Task Descriptions in In-Context Learning

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October 1, 2026

Abstract

In-context learning (ICL) allows models to adapt to tasks using prompts that typically combine a natural language task description with input-output demonstrations. While the informational value of demonstrations is well understood, quantifying the precise contribution of the task description remains challenging. By formulating ICL as Bayesian inference over a latent task, we treat the description and demonstrations as distinct modalities of observations. We quantify the description’s value via its *effective sample size* (ESS)—the equivalent number of demonstrations required to achieve an identical reduction in one-step predictive log-loss regret for a Bayes-optimal predictor. In the exact setting of in-context linear regression, we demonstrate that a description’s ESS is governed by its *precision* (the reduction in prior variance) and its *reliability* (the probability of correct specification). We prove that for a correctly specified description, the ESS is approximately $(d - 1)(1 - r) + 1/(ra_0)$, linear in the task dimension d and in the precision $1/r$. Conversely, for a potentially misspecified description, the ESS is strictly bounded without demonstrations, as the optimal predictor must maintain posterior uncertainty over the description’s validity. Demonstrations act as evidence to resolve this unidentifiability, effectively unlocking the description’s value. Furthermore, we show that small meta-trained Transformers behave Bayes-optimally within the constraints of their output representations; when faced with misspecified descriptions, unimodal output heads force the model to project the optimal bimodal mixture onto a single Gaussian. Finally, we demonstrate that a network’s reliance on the description is inherited from its meta-training distribution, where over-reliance on highly precise but misspecified descriptions yields more regret than ignoring the description.

1 Introduction

The empirical success of large language models is heavily reliant on in-context learning (ICL), where a model adapts to a novel task given a prompt (Brown et al., 2020). Standard prompts incorporate two distinct sources of task information: a natural language task description and a few demonstration examples. In practice the description is an instruction in natural language; we read it as a statement about the task, since its value lies in what it says the task is, and set aside how a model is made to follow it. Empirical observations regarding the description’s utility are heavily mixed; while a well-crafted description can substitute for hundreds of examples (Le Scao and Rush, 2021; Honda et al., 2025), models are also known to perform adequately with misleading descriptions (Webson and Pavlick, 2022) or heavily prioritize demonstrations over explicit instructions (Gupta et al., 2025). Existing theoretical frameworks for ICL often model the prompt strictly as a sequence of demonstrations (Garg et al., 2022; Akyürek et al., 2023; Xie et al., 2022), leaving a fundamental theoretical gap: how can we rigorously quantify the informational equivalent of a task description?

We address this by adopting a Bayesian perspective on ICL (Xie et al., 2022). In this framework, the pretraining distribution constitutes a prior over a latent task. The task description acts as a direct, explicit observation of the task parameters, whereas demonstrations serve as indirect observations. Conditioning on the description yields an updated task prior. In Bayesian statistics, the information contained in a prior can be quantified as its effective sample size (ESS) (Morita et al., 2008), which is dictated by the prior’s concentration (its *precision*) and the probability of prior-data conflict (its *reliability*) (Evans and Moshonov, 2006).

We analyze this dynamic within noisy in-context linear regression with a Gaussian task prior (Garg et al., 2022; Akyürek et al., 2023; Raventós et al., 2023), where Bayes-optimal inference admits closed-form solutions. This tractability allows us to exactly compute a description’s ESS and provides exact reference predictors to evaluate the behavior of meta-trained Transformer models (Genewein et al., 2025).

2 Related Work

Our approach builds upon the understanding of ICL as implicit Bayesian inference (Xie et al., 2022), where meta-trained Transformers have been shown to implement optimal estimators such as Bayes-optimal ridge regression (Akyürek et al., 2023; Raventós et al., 2023; Panwar et al., 2024). While some literature has integrated task descriptions—modeling them as tokens that indicate hypothesis classes or carry input means (Huang and Ge, 2025; Lin et al., 2025; Tong et al., 2026; Lin and Lee, 2024)—these works generally do not quantify the description’s value in units of examples, nor do they formalize the probability of description misspecification. Closest to our setting, Zhu et al. (2026) show that a Transformer given a prefix of related datasets performs amortised hierarchical Bayesian prediction; we express the value of such conditioning information in demonstrations and allow it to be misspecified. Our ESS formulation aligns with prior sample size methodologies in Bayesian statistics (Morita et al., 2008; Reimherr et al., 2021; Neuenschwander et al., 2020), particularly those utilizing predictive regret (Reznik, 2026). To model potential misspecification, we employ robust mixture priors, a technique commonly used in clinical trials to integrate potentially conflicting historical data (Schmidli et al., 2014).

3 Problem Formulation

Tasks and Demonstrations. We assume a latent task represented by a weight vector $w \in \mathbb{R}^d$. A demonstration consists of an input $x \sim \mathcal{N}(0, I_d)$ and a label $y = w^\top x + \epsilon$, with observation noise $\epsilon \sim \mathcal{N}(0, \sigma_y^2)$. The base prior over tasks is isotropic Gaussian $w \sim \mathcal{N}(0, s_0^2 I_d)$. The prior signal-to-noise ratio (SNR) is defined as $a_0 = s_0^2 / \sigma_y^2$, which we set to $a_0 = 10$.

Task Descriptions. A description is formalized as a vector $m \in \mathbb{R}^d$. Its generative process depends on a latent Bernoulli indicator $Z \sim \text{Bernoulli}(p)$, which dictates whether the description is correctly specified ($Z = 1$) or misspecified ($Z = 0$). The generative process is:

$$m \sim \mathcal{N}(0, (1 - r)s_0^2 I_d) \quad (1)$$

$$w \mid m, Z = 1 \sim \mathcal{N}(m, rs_0^2 I_d) \quad (2)$$

$$w \mid m, Z = 0 \sim \mathcal{N}(0, s_0^2 I_d) \quad (3)$$

Marginalizing over Z , the prior for w given the description m is the robust mixture $p\mathcal{N}(m, rs_0^2 I_d) + (1 - p)\mathcal{N}(0, s_0^2 I_d)$ (Schmidli et al., 2014). The parameter $r \in (0, 1)$ denotes the description’s **precision** (the ratio of the conditional variance to the base prior variance). The parameter p denotes the description’s **reliability**. We call a description with $p = 1$ *correctly specified* and one with $p < 1$ *potentially misspecified*.

Regret and ESS. We evaluate a predictor’s uncertainty given a prompt D (the description, n demonstrations, and a query x) via its one-step expected excess log-loss regret relative to an oracle that knows the true task w , the per-step regret of Genewein et al. (2025):

$$R = \mathbb{E} [-\log q(y \mid D) + \log \phi(y; w^\top x, \sigma_y^2)] \quad (4)$$

where ϕ is the Gaussian density and $q(y \mid D)$ is the predictor’s predictive density. Let $R^{\text{desc}}(n)$ denote the Bayes-optimal regret with the description and n demonstrations, $R^{\text{ex}}(n)$ the regret with n demonstrations alone, and $R_{p=1}^{\text{desc}}(n)$ the former with a correctly specified description.

Definition 1. *The Effective Sample Size (ESS) of a description, given n demonstrations, is $\text{ESS}_n = n^* - n$, where n^* is the root of $R^{\text{ex}}(n^*) = R^{\text{desc}}(n)$, calculated via linear interpolation. The standalone ESS is ESS_0 .*

This is the data-dependent prior sample size of Reimherr et al. (2021) with predictive regret as the measure of uncertainty; it is negative when the description hurts.

Reference Predictors and Reliance. A learner applies an effective mixture weight q to the description component. An *oracle-calibrated* predictor uses $q = p$ (Bayes-optimal). An *overconfident* predictor assumes $q = 1$. We define a learner’s *implied reliance* as the value q that minimizes the mean squared error between its posterior-mean predictions and those of the corresponding reference model.

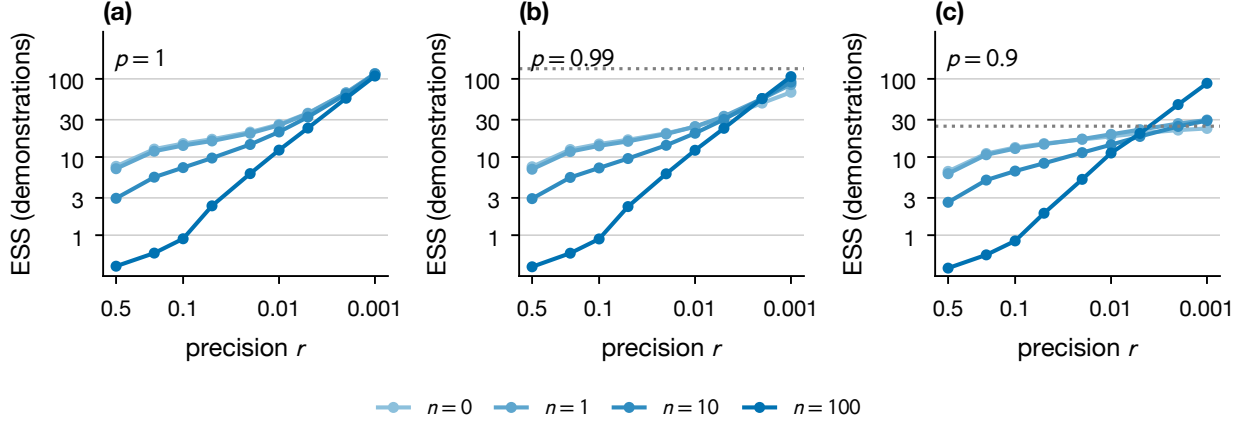


Figure 1: ESS of a description against its precision r with n demonstrations in hand (ESS_n of Definition 1), one panel per reliability p ; $d = 16$, $a_0 = 10$, one seed of 8,000 prompts. Dotted: the limit of the standalone ESS as $r \rightarrow 0$.

4 The Value of Precision (Correctly Specified Descriptions)

When a description has perfect reliability ($p = 1$), the posterior remains Gaussian. The regret is half the expected log of the remaining predictive variance, $R = \frac{1}{2} \mathbb{E} \log(1 + x^\top \Sigma x)$.

Proposition 1. For $p = 1$ and conjugate-prior ESS $c = 1/(ra_0) = \sigma_y^2/(rs_0^2)$:

1. (Coarse descriptions) If $c \leq 1$, then as $rd \rightarrow \infty$, $\text{ESS} = (d-1)(1-r) + (1-r)c + O(1/(rd))$.
2. (Precise descriptions) If $c \geq d$, then as $c/d \rightarrow \infty$, $\text{ESS} = c + d + 1 - 1/a_0 + O(d/c)$.
3. For every d, a_0, r , we have $\text{ESS} \leq c + d + 2$.

Appendix A.1 states the proposition in a fuller form and proves it. Both asymptotic regimes demonstrate that the ESS of a correctly specified description decomposes into a high-SNR term (representing the fraction of the dimension it constrains) and the conjugate-prior ESS c (Neuenschwander et al., 2020). Roughly, $\text{ESS} \approx (d-1)(1-r) + 1/(ra_0)$; on our grid this rule is within about two demonstrations of the exact ESS. At $d = 16$ and $a_0 = 10$ a description that removes nine tenths of the prior variance ($r = 0.1$) is worth 14.9 demonstrations and one that leaves a thousandth ($r = 0.001$) is worth 116 (Figure 1a).

5 The Identifiability Bottleneck (Potentially Misspecified Descriptions)

When $p < 1$, the Bayes-optimal posterior predictive is a two-component mixture. The optimal predictor must maintain uncertainty over the validity of the description.

Proposition 2. Let $\pi_n = P(Z = 1 \mid D_n)$ be the posterior probability that the description is correctly specified given the prompt D_n , and let $\pi_n(Z)$ denote the posterior weight on the true mixture component: π_n if $Z = 1$ and $1 - \pi_n$ otherwise. The regret is bounded by:

$$pR_{p=1}^{\text{desc}}(n) + (1-p)R^{\text{ex}}(n) \leq R^{\text{desc}}(n) \leq pR_{p=1}^{\text{desc}}(n) + (1-p)R^{\text{ex}}(n) + \mathbb{E}[-\log \pi_n(Z)] \quad (5)$$

The identification cost $\mathbb{E}[-\log \pi_n(Z)]$ is the conditional entropy $H(Z \mid D_n)$, which equals $H(p)$ at $n = 0$ and is non-increasing in n (Appendix A.2). Without demonstrations ($n = 0$), the regret is bounded below by $(1-p)R^{\text{ex}}(0)$. Consequently, the ESS strictly saturates at a finite value, preventing arbitrary scaling with precision as $r \rightarrow 0$. At $d = 16$, $a_0 = 10$ and $p = 0.9$ the limit is 25 demonstrations (dotted in Figure 1c): refining r from 0.1 to 0.001 raises the ESS of a correctly specified description from 14.9 to 116, but that of a $p = 0.9$ description only from 13.1 to 23.1.

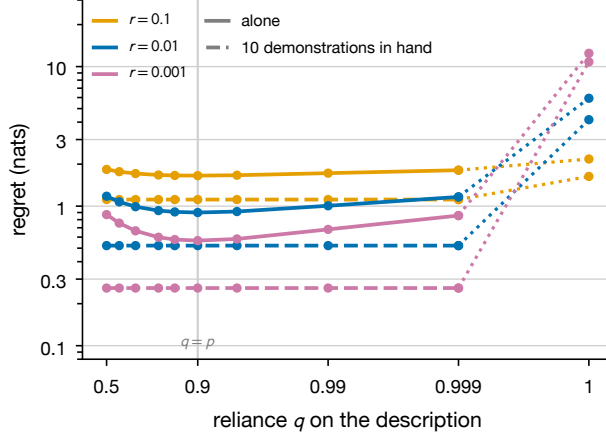


Figure 2: Regret of the Bayes-optimal predictor against the reliance q it places on a description with reliability $p = 0.9$, alone (solid) and with ten demonstrations in hand (dashed); $d = 16$, $a_0 = 10$, one seed of 8,000 prompts.

Demonstrations resolve unidentifiability. As demonstrations accumulate, they provide evidence to infer the latent indicator Z , driving down the conditional entropy $H(Z \mid D_n)$. The ESS of a potentially misspecified description then approaches p times that of a correctly specified one ($\text{ESS}_n \rightarrow pc$). Demonstrations effectively pay down the informational cost of potential misspecification: with 100 demonstrations in hand, the $r = 0.001$, $p = 0.9$ description is worth 88 demonstrations against 108 for a correctly specified one (Figure 1).

The cost of overconfidence. An overconfident predictor ($q = 1$) interacting with a misspecified description incurs severe regret. Because the assumed prior is highly concentrated around the erroneous description, the true generative target falls drastically outside the posterior predictive distribution. For highly precise descriptions, overconfidence is an active liability, yielding higher regret than ignoring the description entirely: at $d = 16$, $a_0 = 10$ and $p = 0.9$, the $r = 0.001$ description alone costs the overconfident predictor 13.4 nats against 2.51 for no description (Figure 2). Under-reliance is cheap by comparison; the regret is flat for a wide band below p , rises gently above it, and jumps at $q = 1$.

6 Empirical Analysis: Meta-Trained Transformers

We meta-trained 12-layer Transformers on sequences generated by our problem formulation ($d = 5$, $a_0 = 10$). The description m is provided via a dedicated prefix token, optimized over Gaussian log-loss.

Setup. Prompts use the prefix embedding of Huang and Ge (2025): an optional description token (their descriptor token) carrying m , up to 20 demonstration tokens each holding an input beside its label, and a query token, with two indicator coordinates marking the token type and no positional encoding. The architecture and optimiser follow Garg et al. (2022): 12 layers, 8 heads, width 256, Adam at a learning rate of 10^{-4} , fresh prompts at every step, 40,000 steps at batch 1024 without a curriculum. The network outputs a mean and a log variance for the query’s label, so its regret is on the scale of (4). The number of demonstrations is uniform on 0 to 20 and the description is omitted in half the prompts. We train one network for each combination of a precise ($r = 0.01$) or coarse ($r = 0.1$) description with a correctly specified ($p = 1$) or potentially misspecified ($p = 0.9$) one, one training run per setting; the description token supplies only m , so each network must learn r and p from its meta-training distribution. We evaluate each network on 20,000 fresh prompts for every n from 0 to 20, with and without a description, on prompts of both reliabilities. Its ESS follows Definition 1 with its own regret curves, and we place it between the reference predictors of §3 on the same prompts (Table 1).

Expressivity bounds Bayes-optimal behavior. On their respective training distributions, Transformers match the Bayes-optimal ESS for perfectly specified descriptions: 14.7 against 15.4 at $r = 0.01$ and 5.24 against 5.28 at $r = 0.1$.

Table 1: Networks against the reference predictors ($d = 5$, $a_0 = 10$, one training run per model, 20,000 evaluation prompts; standard errors of the regrets below 0.03 nats, except 0.10 for the precise $p = 1$ model tested at $p = 0.9$). “Trained” is the Bayes-optimal predictor with reliance p_{train} , “calibrated” the one with reliance p_{test} (where the two differ), and “single Gaussian” the best predictive a mean-and-variance head can express (where the Bayes-optimal predictive is a mixture). The last two columns are the mean regret gap, network minus the trained predictor, over $n = 0, \dots, 20$, with and without a description.

r	p_{train}	p_{test}	ESS				Regret gap (nats)	
			Network	Trained	Calibrated	Single Gaussian	Description	None
0.1	1	1	5.24	5.28	—	—	0.017	0.016
0.1	1	0.9	2.32	2.05	4.26	—	0.010	0.016
0.01	1	1	14.74	15.35	—	—	0.026	0.030
0.01	1	0.9	≤ 0	≤ 0	6.92	—	-0.186	0.030
0.1	0.9	1	4.63	5.05	5.25	—	0.025	0.018
0.1	0.9	0.9	3.57	4.22	—	3.62	0.034	0.018
0.01	0.9	1	6.39	11.90	15.64	—	0.055	0.022
0.01	0.9	0.9	4.30	6.82	—	4.31	0.073	0.022

However, for potentially misspecified descriptions with zero demonstrations, the networks fall short of the exact Bayes-optimal mixture ESS: 4.30 against 6.82 and 3.57 against 4.22. Standard regression heads emit a single mean and variance, inherently lacking the capacity to represent a bimodal posterior mixture. Consequently, the networks project the optimal mixture onto the *best single Gaussian* via moment matching, whose ESS is 4.31 and 3.62. As demonstrations collapse the posterior mixture to a single component, the networks successfully converge with the exact Bayes-optimal predictor: at $r = 0.01$ the network’s regret is within 0.05 nats of it from $n = 6$ on, and its ESS with ten demonstrations in hand is 8.1 against 9.0. The shortfall is not under-training: a model trained at $r = 0.05$, $p = 0.9$ and continued to 80,000 steps kept it (3.9 against 5.0).

Reliance is inherited from meta-training. We demonstrate that the implied reliance of the network is governed by the meta-training distribution. Models trained exclusively on perfect reliability ($p = 1$) learn an implied reliance of $q \approx 1$. At deployment, they act as overconfident predictors: when one description in ten is misspecified, the precise model’s regret with the description is 2.87 nats against 1.87 without it, so its ESS is at most zero where the oracle-calibrated predictor obtains 6.9. Conversely, models trained on $p = 0.9$ learn to maintain structural uncertainty, exhibiting an implied reliance of $q \approx 0.85$ even when evaluated exclusively on correctly specified descriptions, where their ESS is 6.4 against the oracle-calibrated 15.6 at $r = 0.01$.

7 Discussion and Limitations

Our theoretical framework establishes that the ESS of a valid task description scales with its precision and constrained dimensionality. However, potential misspecification introduces an identifiability bottleneck, forcing optimal predictors to maintain uncertainty and capping the description’s value. Demonstrations are critical not just as additive data, but as verification signals that resolve this unidentifiability. Meta-trained Transformers approximate this Bayes-optimal reasoning, bounded by the expressivity of their unimodal output representations, with their propensity for reliance inherited from training.

Our analysis is subject to limitations, notably the assumption of a linear-Gaussian generative process; whether our claims carry over to large language models reading text is open, and the network results rest on one training run per setting. Furthermore, our ESS metric is defined via one-step prediction; under sequential prediction horizons where the predictor’s own feedback accumulates over time, the ESS is lower (7.8 against 14.9 demonstrations at $d = 16$, $a_0 = 10$, $r = 0.1$), representing a valuable direction for future research.

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A Mathematical Proofs

A.1 Proof of Proposition 1 (Correctly Specified Descriptions)

We prove Proposition 1 using the exact regret expansions for the predictive log-loss. Let ψ denote the digamma function, and let $\varepsilon = 1/a_0$. Given n examples $X_n \in \mathbb{R}^{n \times d}$, the posterior covariance is $\Sigma = \sigma_y^2(\varepsilon I + X_n^\top X_n)^{-1}$. For a query $x \sim \mathcal{N}(0, I_d)$, let $A = \|x\|^2 \sim \chi_d^2$. The one-step expected regret is $R^{\text{ex}}(n) = \frac{1}{2} \mathbb{E} \log(1 + x^\top \Sigma x)$.

Regret Expansions. Using properties of Wishart distributions and the Sherman-Morrison formula, we can bound the regret in two regimes:

1. **Fewer than d examples ($n < d$):** Let $k = d - n$. The regret is lower bounded by:

$$2R^{\text{ex}}(n) \geq \log(2a_0) + \psi(k/2) \quad (6)$$

with equality as $a_0 \rightarrow \infty$.

2. **At least d examples ($n \geq d$):** Let $m = n - d + 1$. The regret is upper bounded by:

$$2R^{\text{ex}}(n) \leq \psi\left(\frac{n+1}{2}\right) - \psi\left(\frac{m}{2}\right) \quad (7)$$

Locating the ESS Root. By Definition 1, the ESS is the root n^* of $R^{\text{ex}}(n^*) = R^{\text{desc}}(0)$. For a correctly specified description, $R^{\text{desc}}(0) = \frac{1}{2} \mathbb{E} \log(1 + c^{-1}A)$ where $c = 1/(ra_0)$. Matching the regret expansions to $R^{\text{desc}}(0)$ and applying the asymptotic expansions of the digamma function ($\psi(z) = \log z - 1/(2z) + O(z^{-2})$) yields the roots:

- In the coarse regime ($c \leq 1$), matching the lower bound gives $d - n \approx rd$, leading to $\text{ESS} \approx (d - 1)(1 - r) + (1 - r)c$.
- In the precise regime ($c \geq d$), matching the upper bound gives $n \approx c + d + 1$, leading to $\text{ESS} \approx c + d + 1 - 1/a_0$.

By concavity of the log-loss, the ESS is at most $c + d + 2$ in every regime.

The remainder of this section makes the sketch rigorous: it states the proposition in a fuller form, with the regret expansions behind (1) and (2) and their error terms, proves the two bounds above as lemmas with their expansions, and locates the root.

Proposition 1 (full statement). *Let $p = 1$, let ψ be the digamma function, and let $c = 1/(ra_0) = \sigma_y^2/s_\ell^2$ with $s_\ell^2 = rs_0^2$: the ESS of the description's prior $\mathcal{N}(m, s_\ell^2 I_d)$ in the conjugate sense, noise variance over prior variance (Neuenschwander et al., 2020), the conjugate-prior ESS.*

- (i) Coarse descriptions. Suppose $c \leq 1$. Then for every $n < d$, $R^{\text{ex}}(n) \geq \frac{1}{2} [\log(2a_0) + \psi(\frac{d-n}{2})]$, with equality in the limit $a_0 \rightarrow \infty$, and as $rd \rightarrow \infty$, uniformly in $c \leq 1$,

$$\text{ESS} = (d - 1)(1 - r) + (1 - r)c + O\left(\frac{1}{rd}\right).$$

- (ii) Precise descriptions. Suppose $c \geq d$. Then for every $n \geq d$, $R^{\text{ex}}(n) \leq \Phi(n) := \frac{1}{2} [\psi(\frac{n+1}{2}) - \psi(\frac{n-d+1}{2})]$, with equality in the limit $a_0 \rightarrow \infty$, and as $c/d \rightarrow \infty$, uniformly in $a_0 \geq 1$,

$$\text{ESS} = c + d + 1 - \frac{1}{a_0} + O\left(\frac{d}{c}\right).$$

- (iii) For every d, a_0 , and r , $\text{ESS} \leq c + d + 2$.

Standard facts. Throughout, $\varepsilon = 1/a_0$, $A = \|x\|^2 \sim \chi_d^2$ for the query x , $\Sigma = (\varepsilon I + X_n^\top X_n)^{-1}$, and $g(t) = \mathbb{E} \log(1 + At)$, which is increasing and concave with

$$d(1 - (d + 2)t) \leq g'(t) = \mathbb{E} \frac{A}{1 + At} \leq d, \quad -(d^2 + 2d) \leq g''(t) \leq 0. \quad (8)$$

We use (a) $\mathbb{E} \log \chi_k^2 = \psi(k/2) + \log 2$, $\mathbb{E}[\chi_k^{-2j}] = \prod_{i=1}^j (k - 2i)^{-1}$ for $k > 2j$, and $\psi(z) = \log z - 1/(2z) + O(z^{-2})$; (b) for $W \sim W_q(p, I)$ with $p \geq q$ and $z \sim \mathcal{N}(0, I_q)$ independent of W , $z^\top W^{-1} z$ is distributed as $\chi_q^2 / \chi_{p-q+1}^2$ with independent numerator and denominator (Muirhead, 1982, Theorem 3.2.12), and $\mathbb{E} \text{tr} W^{-2} = q(p - 1)/((p - q)(p - q - 1)(p - q - 3))$ for $p - q \geq 4$ (von Rosen, 1988); (c) $y - y^2/2 \leq \log(1 + y) \leq y$ for $y \geq 0$.

Regret. The posterior predictive is the true conditional law of y , a Gaussian with variance $1 + x^\top \Sigma x$, so its expected log loss is its entropy and $R^{\text{ex}}(n) = \frac{1}{2} \mathbb{E} \log(1 + x^\top \Sigma x)$, as in §4. It is strictly decreasing in n , since by the Sherman–Morrison formula each new demonstration lowers $x^\top \Sigma x$ almost surely. For the description, $n = 0$ and $\Sigma = r a_0 I = I/c$, so by (a) and (c) with $y = c/A$,

$$2R^{\text{desc}} = g(1/c) = \log(2ra_0) + \psi(d/2) + \delta^{\text{desc}}, \quad \delta^{\text{desc}} = \mathbb{E} \log(1 + c/A) = \frac{c}{d-2} + O\left(\frac{c^2}{d^2}\right) \quad (d \geq 5). \quad (9)$$

Locating the root. Let $h(n) = 2R^{\text{ex}}(n) - 2R^{\text{desc}}$ on integers $n \geq 0$, with linear interpolant $\bar{h}$. Then ESS is the unique root of $\bar{h}$, and $h(0) = g(a_0) - g(ra_0) > 0$. In (i) and (ii) we show, for some $n_0 \geq 0$, $\alpha > 0$, and $\theta \rightarrow 0$ along the stated limit,

$$h(n) = \alpha(n_0 - n) + O(\alpha\theta) \quad \text{uniformly over integers } n \geq 0 \text{ with } |n - n_0| \leq 3. \quad (10)$$

Being a convex combination of two such values, $\bar{h}(t)$ obeys the same expansion at real $t \geq 0$ with $|t - n_0| \leq 2$; so for small θ the root lies within 2 of n_0 , and $0 = \bar{h}(\text{ESS}) = \alpha(n_0 - \text{ESS}) + O(\alpha\theta)$ gives $\text{ESS} = n_0 + O(\theta)$.

Lemma A.1 (fewer than d demonstrations). *Let $n < d$ and $k = d - n$. Then $2R^{\text{ex}}(n) = \log(2a_0) + \psi(k/2) + \delta^{\text{ex}}$ with $\delta^{\text{ex}} \geq 0$, $\delta^{\text{ex}} \rightarrow 0$ as $a_0 \rightarrow \infty$, and, for $k \geq 5$,*

$$\delta^{\text{ex}} = \frac{\varepsilon(d-1)}{(k-1)(k-2)} + O\left(\frac{\varepsilon^2 d^2}{k^4}\right).$$

Proof. Write $X_n^\top X_n = \sum_{i \leq n} \lambda_i u_i u_i^\top$ with $\lambda_i > 0$ almost surely, and let Q be the squared norm of the projection of x onto the k directions orthogonal to the u_i . Then Σ has eigenvalue a_0 on those directions and $1/(\varepsilon + \lambda_i)$ on u_i , so

$$1 + x^\top \Sigma x = a_0 Q + T, \quad T = 1 + V, \quad V = \sum_{i \leq n} \frac{z_i^2}{\varepsilon + \lambda_i}, \quad z_i = u_i^\top x,$$

where, given X_n , the z_i are i.i.d. standard normal and independent of $Q \sim \chi_k^2$. Taking logarithms and expectations, by (a),

$$2R^{\text{ex}}(n) = \log(2a_0) + \psi(k/2) + \delta^{\text{ex}}, \quad \delta^{\text{ex}} = \mathbb{E} \log\left(1 + \frac{\varepsilon T}{Q}\right) \geq 0.$$

As $a_0 \rightarrow \infty$, εT decreases to 0 pathwise, so $\delta^{\text{ex}} \rightarrow 0$ by dominated convergence (the integrand at $a_0 = 1$ is integrable).

For the expansion, let $F = \sum_i z_i^2 / \lambda_i$. In the eigenbasis of $W = X_n X_n^\top \sim W_n(d, I)$, which has the same eigenvalues, $F = z^\top W^{-1} z$ and $\sum_i z_i^2 / \lambda_i^2 = z^\top W^{-2} z$ with $z \sim \mathcal{N}(0, I_n)$ independent of W . By (b), $F \sim \chi_n^2 / \chi_{k+1}^2$, so $\mathbb{E} F = n/(k-1)$ and $\mathbb{E}(1+F)^2 = O(d^2/k^2)$, and $\mathbb{E} \sum_i z_i^2 / \lambda_i^2 = n(d-1)/(k(k-1)(k-3))$. Since $1/\lambda - \varepsilon/\lambda^2 \leq 1/(\varepsilon + \lambda) \leq 1/\lambda$, we have $F - \varepsilon \sum_i z_i^2 / \lambda_i^2 \leq V \leq F$. Now (c) with $y = \varepsilon T/Q$, the independence of Q from T , and (a) give

$$\frac{\varepsilon(1 + \mathbb{E} V)}{k-2} - \frac{\varepsilon^2 \mathbb{E} T^2}{2(k-2)(k-4)} \leq \delta^{\text{ex}} \leq \frac{\varepsilon(1 + \mathbb{E} F)}{k-2} = \frac{\varepsilon(d-1)}{(k-1)(k-2)},$$

and the two sides differ by $\varepsilon^2 \mathbb{E} \sum_i z_i^2 / \lambda_i^2 / (k-2) + \varepsilon^2 \mathbb{E} T^2 / (2(k-2)(k-4)) = O(\varepsilon^2 d^2 / k^4)$, as $\mathbb{E} T^2 \leq \mathbb{E}(1+F)^2$. $\square$

Lemma A.2 (at least d demonstrations). *Let $n \geq d$ and $m = n - d + 1$. Then $2\Phi(n) = \mathbb{E} \log(1 + A/Q)$ with $Q \sim \chi_m^2$ independent of A , and $R^{\text{ex}}(n) \leq \Phi(n)$, with equality as $a_0 \rightarrow \infty$. If moreover $a_0 \geq 1$ and $m \geq \max(d, 7)$, then*

$$2R^{\text{ex}}(n) = 2\Phi(n) - \frac{\varepsilon d}{m^2} + O\left(\frac{\varepsilon d^2}{m^3}\right).$$

Proof. Rotate coordinates so that $x = \|x\|e_1$; the rows of X_n remain i.i.d. $\mathcal{N}(0, I_d)$, independent of A . Then $x^\top \Sigma x = A \Sigma_{11} = A/S$ with S the Schur complement of the first coordinate in Σ^{-1} , which the Woodbury identity writes, with X_1 the first column of X_n and X_{-1} the other $d-1$ columns, as

$$S = \varepsilon + X_1^\top (I_n + a_0 X_{-1} X_{-1}^\top)^{-1} X_1.$$

Given X_{-1} , the middle matrix has eigenvalue 1 on the m directions orthogonal to the columns of X_{-1} and $1/(1+a_0\mu_j)$ on the other $d-1$, where μ_j are the eigenvalues of $M = X_{-1}^\top X_{-1} \sim W_{d-1}(n, I)$. Since $X_1 \sim \mathcal{N}(0, I_n)$ is independent of X_{-1} , its coordinates in that eigenbasis are i.i.d. standard normal, and

$$S = Q + \Delta, \quad Q \sim \chi_m^2, \quad \Delta = \varepsilon + \sum_{j < d} \frac{z_j^2}{1 + a_0\mu_j} \in [\varepsilon, \varepsilon(1+F)], \quad F = \sum_{j < d} \frac{z_j^2}{\mu_j},$$

with A, Q , and (z, M) independent; by (b), $F \sim \chi_{d-1}^2/\chi_{m+1}^2$, so $\mathbb{E}F = (d-1)/(m-1)$ and, when $m \geq \max(d, 7)$, $\mathbb{E}(1+F)^2 \leq 5$.

Since $A+Q \sim \chi_{n+1}^2$, (a) gives $\mathbb{E} \log(1+A/Q) = \psi(\frac{n+1}{2}) - \psi(\frac{m}{2}) = 2\Phi(n)$. As $S \geq Q$, $R^{\text{ex}}(n) \leq \Phi(n)$, with equality as $a_0 \rightarrow \infty$ by monotone convergence, since Δ decreases to 0.

For the expansion, $2R^{\text{ex}}(n) = 2\Phi(n) - \mathbb{E}D$ with

$$D = \log\left(1 + \frac{A}{Q}\right) - \log\left(1 + \frac{A}{S}\right) = \log(1+y_1) - \log(1+y_2), \quad y_1 = \frac{\Delta}{Q} \geq y_2 = \frac{\Delta}{Q+A}, \quad y_1 - y_2 = \frac{\Delta A}{Q(Q+A)}.$$

By concavity of $\log(1+y)$ and $1/(1+y_1) \geq 1-y_1$, $(y_1-y_2)(1-\Delta/Q) \leq D \leq y_1 - y_2$. Here Δ is independent of (A, Q) ; $\mathbb{E} \frac{A}{Q(Q+A)} = \mathbb{E} \frac{1}{Q} - \mathbb{E} \frac{1}{Q+A} = \frac{d}{(m-2)(m+d-2)}$ by (a), as $Q+A \sim \chi_{m+d}^2$; $\frac{A}{Q^2(Q+A)} \leq \frac{A}{Q^3}$; and $\varepsilon \leq \mathbb{E}\Delta \leq \varepsilon(1+\mathbb{E}F) = \varepsilon \frac{m+d-2}{m-1}$, $\mathbb{E}\Delta^2 \leq 5\varepsilon^2$. Hence

$$\frac{\varepsilon d}{(m-2)(m+d-2)} - \frac{5\varepsilon^2 d}{(m-2)(m-4)(m-6)} \leq \mathbb{E}D \leq \frac{\varepsilon d}{(m-1)(m-2)},$$

and both sides are $\varepsilon d/m^2 + O(\varepsilon d^2/m^3)$, using $\varepsilon \leq 1$. $\square$

Proof of Proposition 1. (iii). Conditionally on A , $\log(1+Av)$ is concave in v , so Jensen's inequality with $\mathbb{E}[1/Q] = 1/(m-2)$ gives $2\Phi(n) \leq g(1/(m-2))$ for $m \geq 3$. If $m-2 \geq c$, that is $n \geq c+d+1$, then $g(1/(m-2)) \leq g(1/c) = 2R^{\text{desc}}$, so $R^{\text{ex}}(n) \leq \Phi(n) \leq R^{\text{desc}}$ by Lemma A.2. The least such integer is $\lceil c+d+1 \rceil < c+d+2$, and R^{ex} is decreasing, so $\text{ESS} \leq c+d+2$.

(i). The lower bound and the limit are Lemma A.1. For the expansion let $c \leq 1$, $n_0 = (d-1)(1-r) + (1-r)c$, and $n \geq 0$ an integer with $|n-n_0| \leq 3$; write $k = d-n = rd+u$. Since $n_0 = d-rd-u_0$ with $u_0 = 1-r+\varepsilon-c \in [-c, 1]$, we have $|u| \leq 4$, so $k \geq 5$ once $rd \geq 9$. By Lemma A.1, (9), $\psi(z) = \log z - 1/(2z) + O(z^{-2})$, and $\varepsilon = rc$, with every O uniform in $c \leq 1$ and $|u| \leq 4$,

$$\begin{aligned} h(n) &= \psi(k/2) - \psi(d/2) - \log r + \delta^{\text{ex}} - \delta^{\text{desc}} \\ &= \left[\log \frac{k}{rd} - \frac{1}{k} + \frac{1}{d} \right] + \frac{\varepsilon d}{r^2 d^2} \left(1 + O\left(\frac{1}{rd}\right) \right) - \frac{c}{d} + O\left(\frac{1}{(rd)^2}\right) = \frac{u-1+r+c-\varepsilon}{rd} + O\left(\frac{1}{(rd)^2}\right), \end{aligned}$$

since $\log(k/(rd)) = u/(rd) + O((rd)^{-2})$, $1/k = 1/(rd) + O((rd)^{-2})$, $\varepsilon d/(r^2 d^2) = c/(rd)$, and $c/d = \varepsilon/(rd)$. So $h(n) = \frac{1}{rd}[(n_0 - n) + O(1/(rd))]$, which is (10) with $\alpha = \theta = 1/(rd)$, and $\text{ESS} = (d-1)(1-r) + (1-r)c + O(1/(rd))$.

(ii). The upper bound and the limit are Lemma A.2. For the expansion let $n_0 = c+d+1-\varepsilon$ and n an integer with $|n-n_0| \leq 3$, so that $m-2 = c+s$ with $s = n-n_0-\varepsilon$, $|s| \leq 4$; for $c \geq 9d$ every such n has $m \geq \max(d, 7)$. By (9) and Lemma A.2, $2\Phi(n) - 2R^{\text{desc}} = \mathbb{E}[g(1/Q) - g(1/c)]$, and a second-order Taylor expansion of g about $1/c$ with (8), $\mathbb{E}[1/Q] - 1/c = -s/(c(c+s))$, and $\mathbb{E}(1/Q - 1/c)^2 = \text{Var}(1/Q) + s^2/(c(c+s))^2 = O(c^{-3})$ gives

$$2\Phi(n) - 2R^{\text{desc}} = -\frac{s}{c(c+s)} d \left(1 + O\left(\frac{d}{c}\right) \right) + O\left(\frac{d^2}{c^3}\right) = -\frac{ds}{c^2} + O\left(\frac{d^2}{c^3}\right).$$

With Lemma A.2, $\varepsilon \leq 1$, and $\varepsilon d/m^2 = \varepsilon d/c^2 + O(d/c^3)$,

$$h(n) = -\frac{d}{c^2} \left[s + \varepsilon + O\left(\frac{d}{c}\right) \right] = \frac{d}{c^2} \left[(n_0 - n) + O\left(\frac{d}{c}\right) \right],$$

which is (10) with $\alpha = d/c^2$ and $\theta = d/c$, uniformly in $a_0 \geq 1$, so $\text{ESS} = c+d+1-1/a_0 + O(d/c)$. $\square$

A.2 Proof of Proposition 2 (Potentially Misspecified Descriptions)

Let $D_n = (x_{1:n}, y_{1:n}, m, x)$ denote the prompt and the query input. Let $Z \in \{0, 1\}$ be the true validity indicator of the description.

Step 1: The Mixture Predictive. The Bayes-optimal posterior predictive is a π_n -mixture of the correctly specified predictive f_1 and the base prior predictive f_0 , where $\pi_n = P(Z = 1 \mid D_n)$. That is, $q(y) = \pi_n f_1(y) + (1 - \pi_n) f_0(y)$, where f_1 and f_0 are the predictives of the two components given D_n : f_1 is the correctly specified predictor's, and f_0 is the demonstrations-only predictor's, because under $Z = 0$ the description is independent of w and carries no information about y . At $n = 0$, $\pi_0 = p$: seeing m says nothing about Z , because m has the same law $\mathcal{N}(0, (1 - r)a_0 I)$ either way, and x is independent of Z , so there is nothing to update on. The components are then $f_1 = \mathcal{N}(m^\top x, 1 + ra_0 \|x\|^2)$ and $f_0 = \mathcal{N}(0, 1 + a_0 \|x\|^2)$.

Step 2: The Oracle Twin Lower Bound. Consider an oracle predictor that is explicitly given Z . This twin predicts using f_1 when $Z = 1$ (incurring regret $R_{p=1}^{\text{desc}}(n)$) and f_0 when $Z = 0$ (incurring regret $R^{\text{ex}}(n)$). The twin's expected regret is exactly $pR_{p=1}^{\text{desc}}(n) + (1 - p)R^{\text{ex}}(n)$. Because additional information cannot hurt a Bayes-optimal predictor, the expected log-loss of our mixture predictor q cannot be lower than the oracle twin's log-loss. Explicitly, given (D_n, Z) the true conditional density of y is f_Z , and the expected log loss of any density q exceeds that of f_Z by $\mathbb{E}[-\log q \mid D_n, Z] - \mathbb{E}[-\log f_Z \mid D_n, Z] = \text{KL}(f_Z \parallel q) \geq 0$; averaging over Z and D_n gives the bound. Thus:

$$R^{\text{desc}}(n) \geq pR_{p=1}^{\text{desc}}(n) + (1 - p)R^{\text{ex}}(n) \quad (11)$$

Step 3: The Identification Cost Upper Bound. Pointwise, the mixture density satisfies $q \geq \pi_n f_1$ and $q \geq (1 - \pi_n) f_0$. Consequently:

$$-\log q \leq -\log f_1 - \log \pi_n \quad (12)$$

$$-\log q \leq -\log f_0 - \log(1 - \pi_n) \quad (13)$$

Taking the expectation over the true data generating process f_Z and the latent variable Z yields the upper bound. The penalty term simplifies to the conditional entropy $\mathbb{E}[-\log \pi_n(Z)] = H(Z \mid D_n)$. At $n = 0$ it is $p \log(1/p) + (1 - p) \log(1/(1 - p)) = H(p)$.

Step 4: Monotonicity and Saturation. The conditional entropy $H(Z \mid D_n)$ is non-increasing in n , since conditioning on additional demonstrations can only reduce entropy. At $n = 0$, $H(Z \mid D_0) = H(p)$. In detail, the query x is independent of Z and of everything else in D_n , so $H(Z \mid D_n) = H(Z \mid x_{1:n}, y_{1:n}, m)$, and these conditioning sets are nested in n . Evaluating the lower bound at $n = 0$ gives $R^{\text{desc}}(0) \geq (1 - p)R^{\text{ex}}(0)$. Since $R^{\text{ex}}(n)$ is strictly decreasing in n , there exists a finite n^* such that $R^{\text{ex}}(n^*) = (1 - p)R^{\text{ex}}(0)$. The ESS cannot exceed this n^* , so it saturates. The lower bound at $n = 0$ holds for every r because $R_{p=1}^{\text{desc}}(0) = \frac{1}{2} \mathbb{E} \log(1 + ra_0 \|x\|^2)$ is nonnegative; and $R^{\text{ex}}(n)$ is non-increasing because a predictor given $n + 1$ demonstrations can ignore one, and Bayes-optimal regret cannot rise with more information. Finally $R_{p=1}^{\text{desc}}(0)$ decreases to 0 as $r \rightarrow 0$ by monotone convergence, which gives the two limits of the standalone regret, $(1 - p)R^{\text{ex}}(0)$ and $(1 - p)R^{\text{ex}}(0) + H(p)$.